

1 Repository overview (Q&A)

1.1 Q: What is this repository?

A: This is a PyTorch research codebase for **video watch-time prediction**, implementing the EGMN paper and several baselines, with preprocessing and experiments primarily for the public KuaiRec dataset.

1.2 Q: What are the main folders and entrypoints?

A: Models live in `model/`, dataset loading is in `dataloader/`, KuaiRec preprocessing is in `dataset/kuaiRec/`, runnable experiments are in `run_*.py`, and shared metrics/utilities are in `utils.py`.

1.3 Q: What metrics are reported by the run scripts?

A: The run scripts evaluate on `test` and report **MAE** (absolute error), **XAUC** (ordering agreement after sorting by predicted score), and **KL** divergence between binned empirical distributions.

1.4 Q: How do you run EGMN and baselines (example commands)?

A:

```
# Train + evaluate the proposed EGMN model on KuaiRec
python run_egmn.py --dataset_name kuaiRec --device cuda:0

# Baselines (examples)
python run_cread.py --dataset_name kuaiRec
python run_d2q.py --dataset_name kuaiRec
```

1.5 Q: What does the dataloader yield to the model?

A:

```
# Pseudocode reflecting dataloader/kuaiRec.py
for features, label in dataloaders["train"]:
    # features: dict[str, Tensor], label: normalized play_time
    y_hat = model.predict(features)
```

1.6 Q: What do the evaluation helper functions look like (schematically)?

A:

```
# utils.py (simplified)
def eval_mae(labels, scores):
    return np.mean(np.abs(labels - scores))

def eval_xauc(labels, preds):
    # ranks by prediction, then measures label ordering agreement
    ...

def eval_kl(samples_p, samples_q, bins=100, epsilon=1e-10):
    # binned empirical KL divergence
    ...
```